Applying Performance Optimization Techniques (Indexing, Subforms and Query Tuning) to a Zoho Creator Application

Title

Applying performance optimization techniques (indexing, subforms and query tuning) to the Insurance Claim Request & Processing Management application in Zoho Creator.

Aim

To study how indexing, subforms and query tuning help an application run faster and keep its data neat, and to apply these ideas to the Insurance Claim Request & Processing Management application in Zoho Creator.

Objectives

- To find the fields that are searched most often and understand how indexing helps.
- To study how a subform stores many related rows inside one record.
- To read a Deluge script and understand how to fetch only the data that is needed.
- To check the form design and the report of the application.

Software / Tools Required

- Zoho Creator
- Insurance Claim Request & Processing Management application
- Deluge scripting (Form Workflow)
- Web browser with internet connection

Introduction / Theory

Performance optimization: When an application has more data, pages and reports can become slow. Performance optimization means making small design choices so that the app stays quick and easy to manage.

Indexing: An index works like the index of a book. It helps the system find a record without reading every row. Fields that are often used for search, such as Policy_Number or Claim_Type, are good fields to search by.

Subform: A subform is a small table inside a form. It saves many related rows in one record. For example, one claim can have many supporting documents, so a subform avoids repeating the claimant details again and again.

Query tuning: This means asking the database only for what is needed. In Deluge, we use criteria to fetch only the matching records instead of all records. This reduces extra work and saves time.

Procedure

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Step 1: Study the form design

1. Open the Insurance Claim Request & Processing Management application in edit mode.
2. Open the Claim_Request_Form. The fields are grouped into three sections: Claimant Details, Contact Details and Claim Details.
3. Click a field to see its properties. Claimant_Name is a Name field, and it is set as Mandatory.

Step 2: Study the report and indexing

4. Open the Claim_Request_Form Report. It shows 6 records with columns such as Claimant_Name, Claimant_Type, Policy_Number and Claim_Type.
5. Use the search option of the report to find records by a field value. Fields like Policy_Number and Claim_Type are searched often, so they are the best fields to consider for indexing.

Step 3: Study subforms

6. In the form builder, the Advanced Fields list has a SubForm option.
7. A subform such as Supporting_Documents (with fields like Document_Name and Document_Type) can be added to keep many documents under one claim. This is a design idea for the application and was studied here as a concept.

Step 4: Apply query tuning with Deluge

8. Open the form workflow: Claim_Request_Form > Created > Successful form submission > Deluge Claim Processing.
9. Write a script that uses the submitted values directly, without fetching extra records.
10. Click Update to save the script, and submit a test claim to see the output.

Deluge script used

```
info "===== CLAIM PROCESSING TEST =====";
info "Claimant Name: " + input.Claimant_Name;
info "Policy Number: " + input.Policy_Number;
info "Claim Type: " + input.Claim_Type;
info "Claim Amount Requested: " + input.Claim_Amount_Requested;
info "===== END OF TEST =====";
```

Example of a tuned query (for understanding): to get only Motor Insurance claims, we can write motor_claims = Claim_Request_Form[Claim_Type == "Motor Insurance"]; This brings back only the matching records and not the whole table.

Output

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1. Indexing

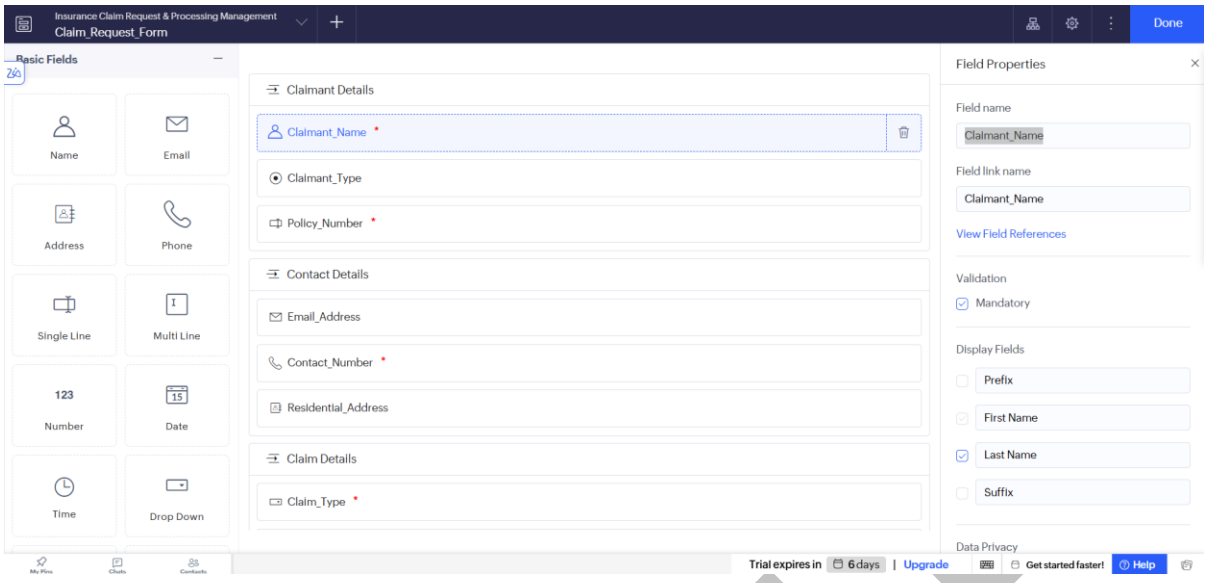


Figure 10.1: Claim_Request_Form with grouped sections and field properties

2. Sub forms

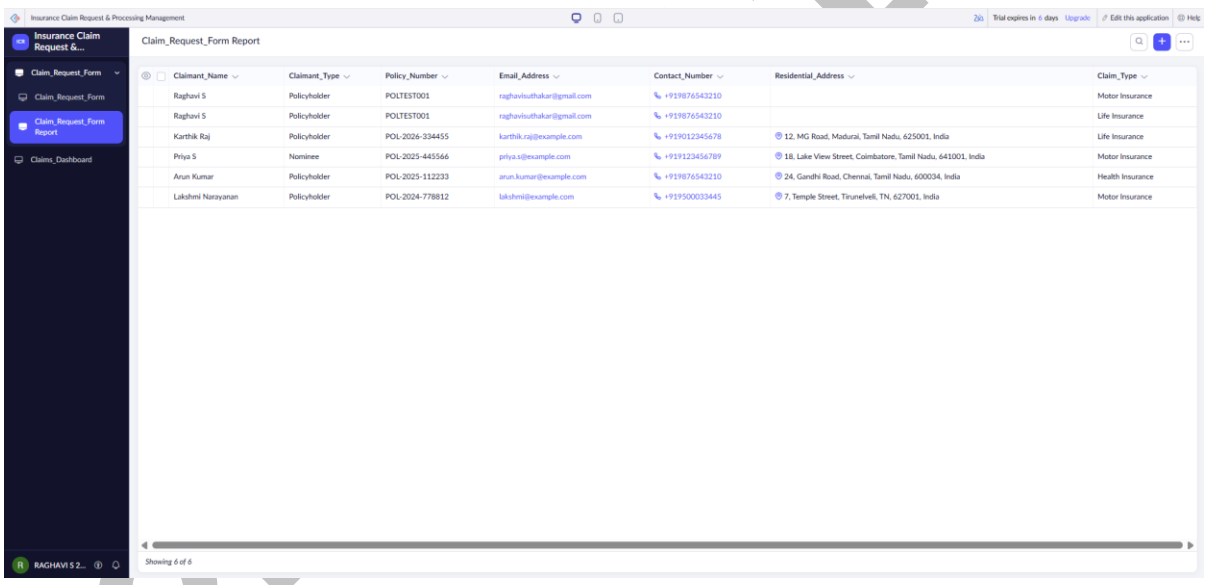


Figure 10.2: Claim_Request_Form Report showing 6 of 6 records

3. Query Tuning

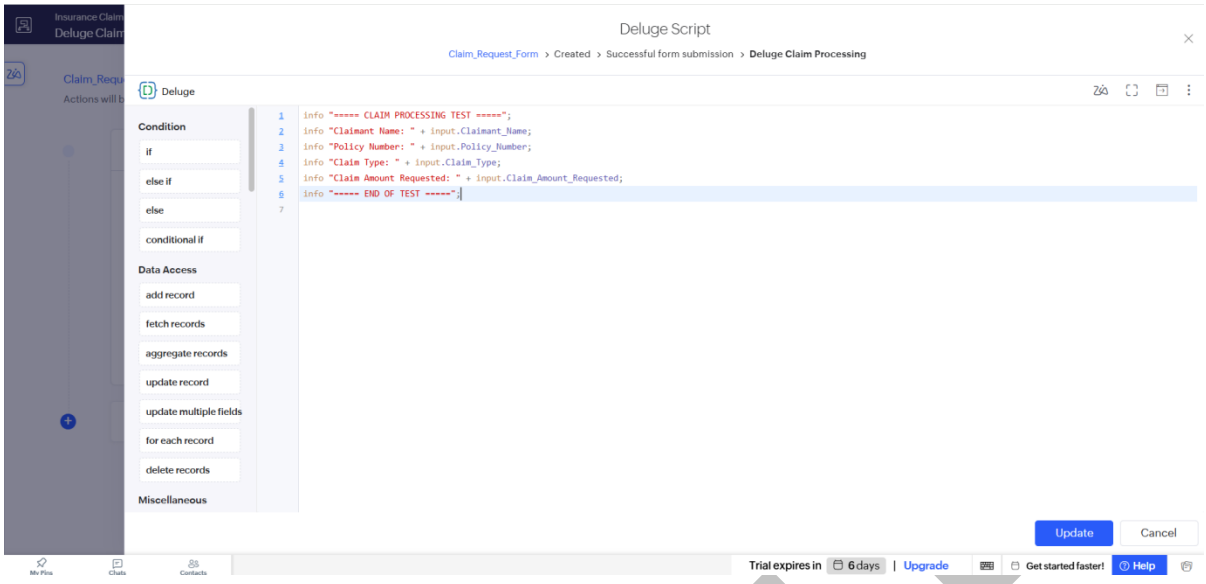


Figure 10.3: Deluge script in the Successful form submission workflow

Optimization Techniques Summary

Technique	What it does	Use in this application
Indexing	Helps find records faster	Search by Policy_Number or Claim_Type
Subform	Stores many related rows in one record	Supporting documents for one claim
Query tuning	Fetches only the needed records	Filter by Claim_Type using criteria

Observation

- The form groups fields into sections, which keeps data entry clear and easy to follow.
- The report shows 6 records. Claim_Type has the values Motor Insurance, Life Insurance and Health Insurance.
- The Deluge script runs after a successful form submission and prints the claimant name, policy number, claim type and requested amount.
- The script uses the submitted input values directly, so it does not read any extra records and stays light.
- The account shows a trial that expires in 6 days, so some features may be limited.

Limitations

Indexing is handled by Zoho Creator inside the platform, so it was studied as a concept and was not changed in this experiment. The subform was also studied as a design idea and is not shown in the screenshots.

Conclusion

Good performance in Zoho Creator comes from small, careful choices. Indexing helps in fast searching, subforms keep related data together in one record, and query tuning makes sure

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only the needed records are read. These steps keep the Insurance Claim application quick as the number of claims grows.

Final Result

- The form design, the report and the Deluge workflow of the application were studied.
- The concepts of indexing, subforms and query tuning were understood and linked to the Insurance Claim application.
- A Deluge script was saved in the Successful form submission workflow to read the submitted claim values.

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